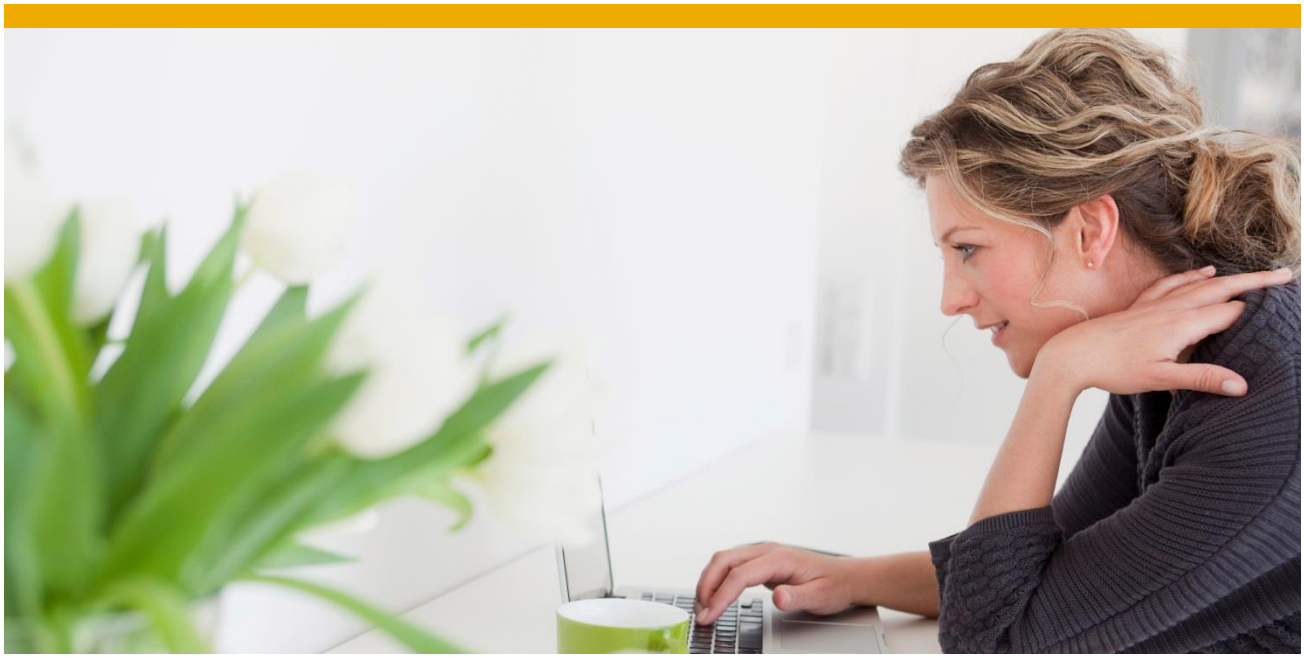

Overview S/4-HA-CLU 1.0 Version 1.03

SAP Application Server
HA Interface Certification
SAP S/4HANA 1809

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ABOUT THE SAP APPLICATION SERVER HA INTERFACE CERTIFICATION

The SAP Application Server HA-Interface-Certification for HA Partners certifies the Start/Stop infrastructure within HA-Setups (SAP NetWeaver Management Agents, Cluster-API).

The overall goal of the HA-Interface-Certification is to establish a SAP HA-Setup scenario that will be known to all HA partners and will be the reference. More specifically for this certification, it is the intention to unify HA-Setups for ABAP platform 1809, which is the underlying technology platform for SAP S/4HANA 1809. In this setup the Standalone Enqueue Server 2 together with the Enqueue Replicator 2 is used. They are available since SAP NetWeaver 7.52 for ABAP.

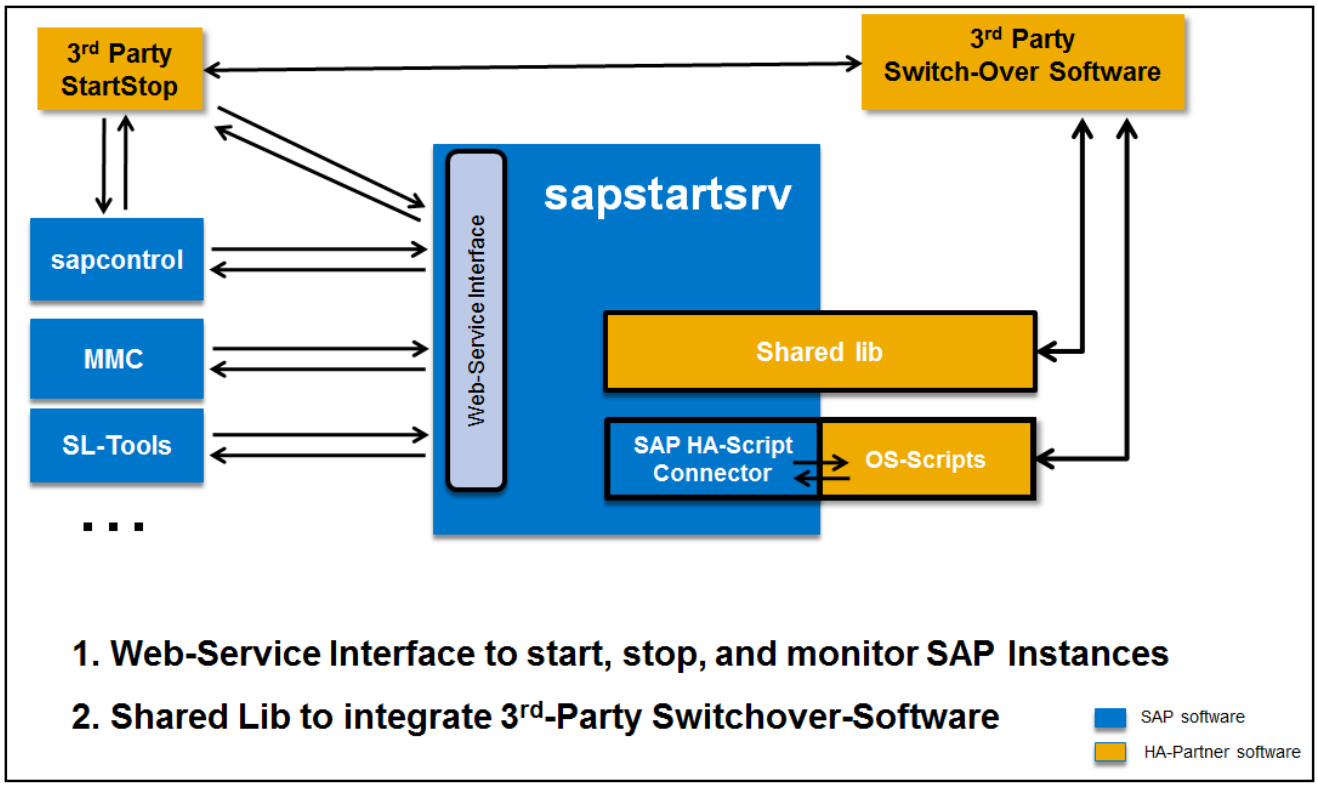
To test the old enqueue mechanism there is a second HA certification named NW-HA-CLU 750 available that uses an HA-Setup for SAP NetWeaver 7.5 for ABAP as well as for Java based application servers. But this certification is not subject of this document.

It is expected to improve the transparency about operating system dependencies and 3rd party failover software as well as to improve the transparency about available SAP HANA-specific HA-Solutions of 3rd party vendors. Additionally, the responsibilities of SAP and its partners, especially for support issues will be clarified.

Important Resources for partners about the HA-Interface Certification can be found here.

Objectives of this certification

The following SAP Start Service (sapstartsrv) interface configuration provides functionality that the HA vendor solution should be able to use in a customer environment:



The following tests will help to verify the integration of the 3rd party software in different situations.

Certification criteria

The following certification criteria will verify the correct functionality and general readiness of the third-party HA software operating the various SAP Software components via the HA-Interface offered by the SAP Start Service (sapstartsrv):

- **General Certification Criteria**
  1. Partner commits to regularly contact SAP for changes in the SAP recommendations for SAP HA-Setup, at the latest before stating support for a new major SAP NetWeaver or SAP S/4HANA release.
  2. SAP-specific documentation is available at partner web site for targeted user groups (at least to basis consultants and customers, if possible, publicly accessible).
  3. Support responsibilities for HA-Setups are defined and agreed in writing within a dedicated SAP note and are released to customers.
  4. Partner informs whether an HA-Setup is offered for SAP HANA database.
- **Technical Certification Criteria**
  1. Name and version of 3rd-party HA software is delivered correctly.
  2. HA configuration is verified successfully.
  3. Successful failover can be performed manually via 3rd-party software.
  4. Crucial SAP-processes are monitored and restored automatically (Restart and/or failover).
  5. Usage of sapstartsrv and integration into 3rd-party failover solution
  6. sapstartsrv status information is integrated into 3rd-party failover detection.
  7. Starting an instance must be possible via standard web service interface (SAPControl).
  8. Stopping an instance must be possible via standard web service interface (SAPControl) without triggering an instance failover.
  9. A document is maintained describing on how to temporarily disable failover software in case of SW maintenance activities. If this is not possible or very difficult, it should describe on how to perform maintenance activities in a fail-over environment. All software maintenance activities that are not possible should be documented as well.

THE CERTIFICATION PROCEDURE IN DETAIL

Preparing the environment

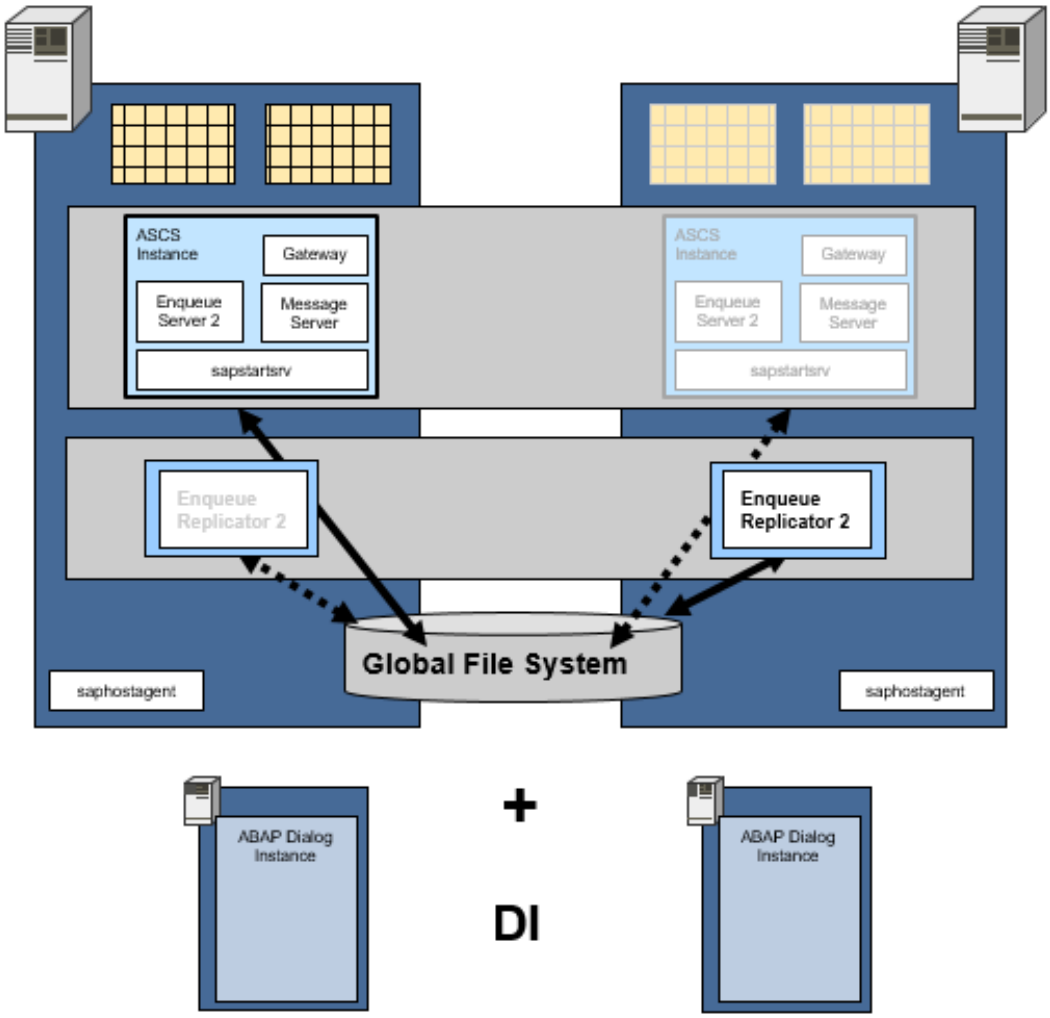
Installing the HA-Solution

The HA-Partner needs to complete all steps required to install the HA-Solution on respective hosts. The HA-Solution must be fully functional and accessible for the certification process.

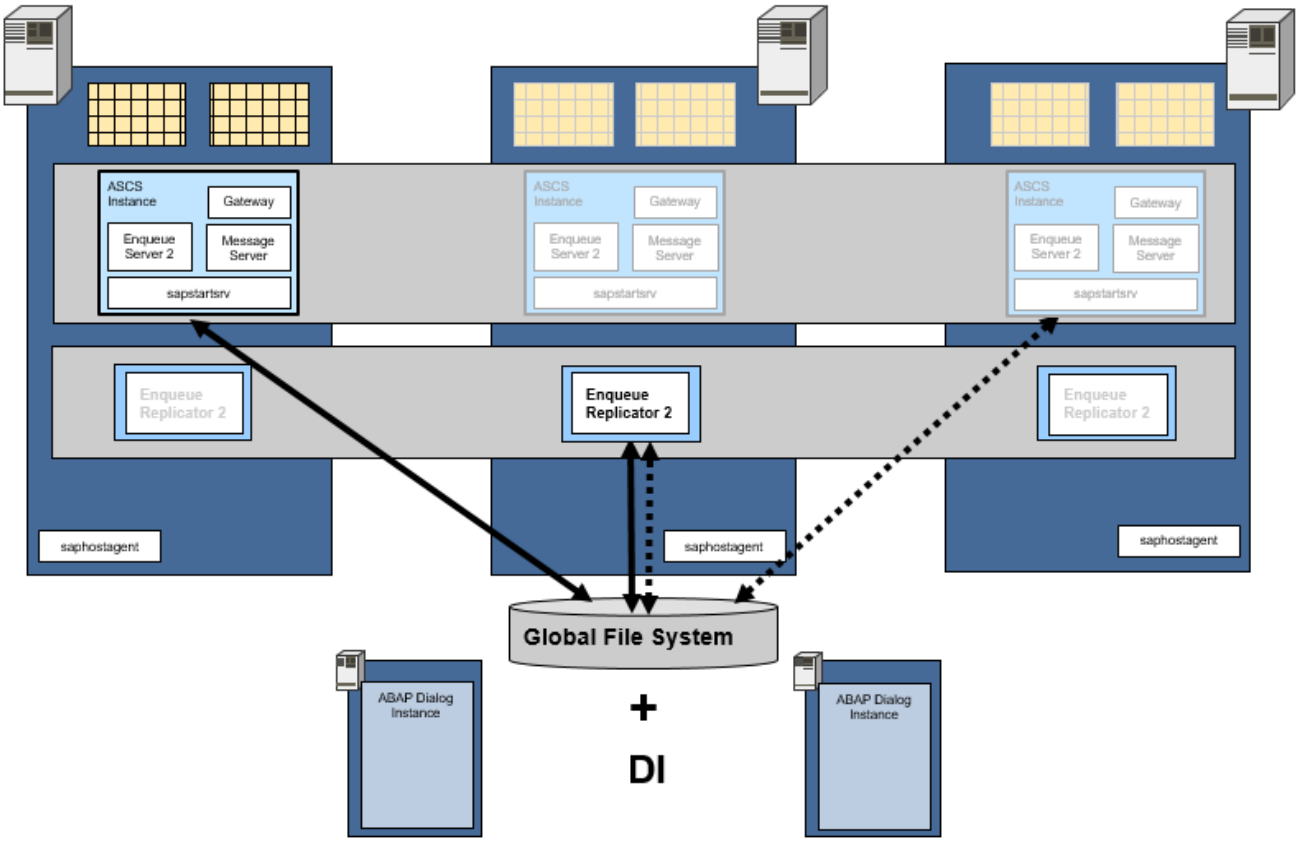
Installing the SAP-System

The recommended behavior in case of failures differs dependent on the number of cluster nodes used for the Standalone Enqueue Server 2 together with the Enqueue Replicator 2. Therefore, two different HA-Setups are the basis of this HA certification: One with 2 cluster nodes and one with more than two cluster nodes. It is necessary to setup both environments with the essential components for the HA certification.

2-Nodes Cluster



n-Nodes Cluster (n > 2)



Before the actual certification process starts, the test environment must be set up by the HA-Partner as follows:

1. An ABAP platform 1809 system must be installed for the certification process. As minimum kernel version SAP Kernel 773 PL 22 + ENSA2_29-xxxxxx.SAR is required. SAP Kernel 773 PL100 and above already contain the ENSA2 patch. Because one of the technical tests requires the exchange of the kernel by a newer one, make sure that you don't start with the newest available kernel patch level.

   The system must be set up by the partner for high availability according to the partner's documentation. A 2-nodes cluster and a cluster with more than two nodes must be provided for executing the certification tests. The test environment can be simulated using virtualization technology.

2. The following SAP components and instances must be installed and setup correctly:
   a. A separated ASCS instance, manageable in a dedicated switch-over group
   b. The Enqueue Replicator 2 (ER2) is setup and manageable in a dedicated switch-over group
   c. The hosts for the ASCS and the ER2 switch-over group should be the same
   d. At least 2 ABAP-Dialog instances are installed on two (virtual or physical) hosts, the hosts of the dialog instances can be used also to hold the ASCS and/or ER2-setup
   e. sapstartsrv is part of the switch-over group and must be moved as well in case of a failover
   f. saphostagent installed once per host and not part of the failover group
   g. Platform dependent SAPCPE for replication of the executables is in place

3. Independently of whether the SAP System is to be integrated into the HA-Solution by using the HA-Interface-Script-Connector or a vendor HA library, the required SAP Kernel patch level must be applied as described in SAP note 1693245.

4. The HA vendor software is up and running and controls the SAP System as desired.
5. An SAP Management Console (SAP MMC or SAP MC) must be installed to monitor the system landscape. All instances must be visible there and should be running properly. To ease the operational tasks, the SAP MMC/MC should be called from a separate machine (frontend machine) to be independent from switching the cluster nodes.

In addition to these mandatorily required setup-characteristics for the HA-Interface certification, the hardware as well as the database is typically set up in a highly available manner (vendor specific) at customer side. To pass the certification, it is not necessary to setup the hardware or database in a highly available manner though, because these components are not in the scope of HA-Interface certification at all.

Installing the HA Test Tool

For some of the tests you need the SAP HA Test Tool. At least version v2.14 must be installed. sapcontrol is needed by the HA Test Tool. It must be installed on the client-side of the tool.

General tests to certify the HA-Solution

Test	Title
GEN01	Recommended SAP HA-Setup is described in vendor specific documentation
GEN02	SAP-specific documentation is available at partner web site
GEN03	Support processes are documented and available for customers
GEN04	Inform whether SAP HANA database is supported by the HA-Solution

Technical tests to certify the HA-Solution

The following tests verify the functional correctness of the HA-Solution in conjunction with the HA-Interface of the SAP Start Service (sapstartsrv).
Each test can be performed several times to verify the repeatability, reliability and completeness of the solution.
If not indicated otherwise, every test can be performed manually using the Web-Service-Interface or using the functionality provided by the HA-Solution, if available.
If not stated otherwise, a test can be executed on one of the two HA setups.

Test	Title	HA Setup (# of nodes)
TEC01	Name and version of 3rd-party HA software is delivered correctly	
TEC02	Verification of HA configuration shows no errors	
TEC03	Shared Lib (HA-Interface) is being loaded without any errors	
TEC04	Manually moving the ASCS works correctly	2
TEC05	Irrecoverable outage of the Enqueue Server 2 is handled correctly	2
TEC06	Outage of the Enqueue Replicator 2 is handled correctly	2
TEC07	ASCS moves correctly under load without specifying the destination node	>2
TEC08	ASCS moves correctly in case of hardware or OS failure	>2
TEC09	Recoverable outage of the Message Server is handled correctly (if the SAP Profile Parameter “Restart_Program” is used for the Message Server)	>2
TEC10	Irrecoverable outage of the Message Server is handled correctly (if the SAP Profile Parameter “Start_Program” is used for the Message Server)	>2
TEC11	Enqueue Replicator 2 moves correctly under load in case of hardware or OS failure	>2
TEC12	Starting and stopping of SAP resources work correctly	
TEC13	Outage / Restart of the SAP Start Service is handled correctly	
TEC14	Automated ASCS restart in place works correctly	
TEC15	Execution of the Rolling Kernel Switch is possible	
TEC16	Simulating an upgrade works correctly using the Maintenance Mode (if the functionality is available)	

TEC17	Simulating an upgrade works correctly (if no Maintenance Mode is available)
TEC18	Simulating cluster failure works correctly (if the functionality is available)

For all test cases, monitoring of the SAP System is crucial, in particular that all instances are available again after the switch-over procedure of an instance. It is of particular interest whether

- the Enqueue lock table is correctly restored (Same entries available in restored table),
- the Enqueue Replication Service is up and running afterwards and the lock table replication is re-established again, and
- the central file systems must be available after the switch again.

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